

MOD_LPC Schema - Current State Analysis

Section 1: Schema at a Glance 2

- 1.1 Database & Schema Identification 2*
- 1.2 Total Schema Size 2*
- 1.3 Total Object Inventory 2*
- 1.4 Table Structure — Partitioned vs Non-Partitioned 2*
- 1.5 LOB Column Inventory 3*

Section 2: Table Analysis & Classification 3

- 2.1 Total Table Count 3*
- 2.2 Classification of Tables 3*
- 2.3 The 34 Functional Groups 5*
- 2.4 Naming Convention Patterns Identified 6*
- 2.5 Empty vs Populated Tables 7*
- 2.6 Tables With No Date Columns — 116 Tables 8*

Section 3: Storage Distribution 9

- 3.1 Storage by Group 9*
- 3.2 Top 10 vs Bottom 10 storage 9*

Section 4: Usage & Activity Metrics 10

- 4.1 Activity by group 10*
- 4.2 Top and Bottom usage metrics 11*
- 4.3 Tables Confirmed Dead — Entire Groups With Zero Activity 11*
- 4.4 Most critical examples — large active tables with stale statistics: 11*

Section 5: Content & Metadata Analysis 13

- 5.1 Capturing comments- Initial analysis for the description of tables 13*
- 5.2 Top 10 widest columns by column count 13*
- 5.3 Table Definitions — Extracted from comments 14*
- 6.1 The following tables were identified as likely pre-2016 candidates 15*

Executive Summary 16

Section 1: Schema at a Glance

1.1 Database & Schema Identification

1. Schema Name: MOD_LPC
2. Database: MOD_PROD

1.2 Total Schema Size

1. Total Storage (TB) : 34.5 TB
2. Total Storage Segments : 2,201- Include everything
3. Storage Cost (Capital) : ~\$70.6 Million at \$2,000/TB

1.3 Total Object Inventory

S.no	Object Type	Count	Description
1	Index Partitions	7,909	Partitioned indexes on partitioned tables
2	Table Partitions	7,859	Individual partition segments of 56 partitioned tables
3	Indexes	1,699	Standard indexes on non-partitioned tables
4	Tables	1,084	Physical tables
5	Types	145	Oracle object type definitions
6	Views	55	Database views
7	Type Bodies	27	Oracle type body implementations
8	LOB Segments	22	Large object columns
9	Queues	12	Oracle Advanced Queuing message queues
10	Sequences	11	Auto-increment number sequences
11	Packages	11	PL/SQL packages
12	Package Bodies	11	PL/SQL package body implementations
13	Rule Sets	10	Oracle rules engine objects
14	Functions	4	PL/SQL standalone functions
15	Evaluation Contexts	3	Oracle rules evaluation contexts
16	Materialized Views	2	Pre-computed result set objects
17	Triggers	1	Database trigger
	Total Objects	~20,925	All schema objects combined

The schema contains 7,859 partitions and 1699 indexes, Oracle backup process each segment independently, That is why backup takes 7 days.

1.4 Table Structure — Partitioned vs Non-Partitioned

Non-Partitioned Tables : 1,028

Partitoned Tables : 56

Total : 1,084

Details of Partitioned Tables

Partitioned Tables : 56

Total Partition Segments Generated: 7,859

Largest Partitioned Table : MKTPLANPNODE- 2,453 partitions

Second Largest Partition: MKTPLANPNODECONSTRAINT_D - 1,580 partitions

1.5 LOB Column Inventory

LOB (Large Object) columns store unstructured data as XML documents, binary files.

The schema contains 22 LOB column definitions across 9 tables.

All 22 LOB column segments are currently consuming zero storage. The columns exist as schema definitions but contain no data.

Section 2: Table Analysis & Classification

2.1 Total Table Count

S.no	Metric	Value
1	Total Physical Tables	1,084
2	Partitioned Tables	56
3	Non-Partitioned Tables	1,028
4	Tables With Data (rows > 0)	~660
5	Empty Tables (0 rows confirmed)	~424
6	Tables With No Date Columns	116
7	Tables With No Documentation	664

2.2 Classification of Tables

naming conventions were analyzed across the full table list to identify consistent patterns.

1. Prefix Analysis: Every distinct table name prefix is taken to identify groups

2. Suffix Analysis: Every distinct table name suffix is taken to identify groups

No single standardized naming convention exists across the schema. Multiple teams over multiple years created tables with inconsistent naming. This made classification of groups incomplete —

A manual review pass was required to catch non-standard backup table names like

BKP_MKTREASON_OCT232025, MKTPLANPNODE_TEMP_4AUG2025, and date-stamped names like **MKTCASE_05042020.**

2.3 The 34 Functional Groups

S.no	Group	Count	Category
1	MKT — Market Tables	672	Core Production
2	AUDMKT — Audit	80	Audit
3	Date-Stamped Backups	84	Backup Artifact
4	STG — Staging	52	Staging
5	SNP — Snapshot	26	Snapshot
6	LPC — Core App	25	Application
7	ETM — Migration/Transfer	18	Migration
8	AQ\$ — Oracle Internal	18	Oracle System
9	BKP — Backup	15	Backup Artifact
10	TEMP — Temporary	10	Temporary
11	SPIN System	9	SPIN reporting system tables.
12	CONSTRAINT Tables	7	Constraint management tables.
13	EMS System	6	Energy Management System interface tables.
14	OLD Suffix	6	Tables explicitly named with _OLD suffix.
15	MSCS System	5	MSCS system tables.
16	Reference / Config	4	Standalone reference and configuration tables.
17	QT — Queue Tables	4	Oracle queue table artifacts.
18	MIGRATION Tables	4	Data migration artifacts. All confirmed empty.
19	BAK / BK / BKUP Suffix	3	Tables with _BAK, _BK, _BKUP suffixes.
20	GenByFuel	3	Generation by fuel mix summary tables.
21	SMD Export	3	SMD export system tables.
22	Equipment / Outage	3	Equipment and outage management tables.
23	RUPD\$ MV Refresh Log	2	Oracle materialized view refresh log. Do not touch.
24	MLOG\$ MV Log	2	Oracle materialized view log.
25	MV — Materialized Views	2	Pre-computed result set tables.
26	OPS — Operations	2	Operational data tables.
27	REGLOC — Regulatory	2	Regulatory location calculation tables.
28	PD — Price Data	2	Price data tables. Both empty.
29	AUD — Audit	2	Standalone audit tables.
30	CASE Tables	2	Case performance statistics tables. Both empty.
31	MOI Temp	2	MOI system temporary tables.
32	Tool Artifacts	2	TOAD_PLAN_TABLE and STAT\$TAB — developer tool leftovers in production.
33	OTHER	14	Tables that did not match any naming pattern. Require manual review.
	Total	1,084	

2.4 Naming Convention Patterns Identified

During classification, the following naming patterns were identified.

Prefix Patterns

S.no	Prefix	Example	Meaning	Tables
1	MKT	MKTPLANPNODE	Market operations table	672
2	AUDMKT	AUDMKTCASEUNIT	Audit trail of MKT table	80
3	AQ\$	AQ\$_MARKET_QT_H	Oracle AQ internal artifact	18
4	BKP_	BKP_MKTCASE1312025	Explicit backup copy	15
5	STG	STG_MKTPLANPNODECONSTRAINT	Staging table	52
6	SNP	SNPBID	Snapshot table	26
7	ETM	ETMMIGRATIONTABLE	ETM system table	18
8	EMS	EMSZONALDISPATCHRATE	EMS system table	6
9	LPC	LPCBORUNITHRLY	LPC application table	25
10	SPIN	SPINREPORT_T	SPIN reporting table	9
11	MSCS	MSCS_PNODE	MSCS system table	5
12	MIGRATION	MIGRATIONTABLE	Migration artifact	4
13	RUPD\$	RUPD\$_MKTPLAN	MV refresh log — Oracle internal	2
14	MLOG\$	MLOG\$_MKTPLAN	MV log — Oracle internal	2
15	MV_	MV_MKTPLAN	Materialized view	2

Suffix Patterns Inside MKT Tables

This is particularly important. Within the 672 MKT tables, multiple suffix patterns exist that indicate copies of base tables:

S.no	Suffix	Example	Meaning	Implication
1	_D	MKTPLANNODE_D	Delta copy	Maintained in parallel with base table
2	_P	MKTPLANNODE_P	Parallel copy	Parallel version of base table
3	_ARCH	MKTPLANNODECONSTRAINT_ARCH	Archive copy	Historical archive — 2,511 GB
4	_OLD	MKTPLANCONSTRAINT_OLD	Old copy	Previous version preserved manually
5	_BAK	MKTEMERGENCYREGIONDEF_BAK	Backup copy	Manual backup
6	_TEMP	MKTPLANNODECONSTRAINT_TEMP	Temporary	Should not persist
7	_I	MKTPLANUNITSCHEDULE_I	Unknown variant	Empty — purpose unclear
8	(Date)	MKTCASE_05042020	Date-stamped backup	Manual backup with date appended

The presence of _D, _P, _ARCH, _OLD, _BAK and date-stamped variants within MKT tables indicates a pattern of manual table copying that accumulated over years without a consistent naming standard or cleanup process. These variants account for a significant portion of storage within the MKT group.

Non-Standard Backup Names — The Problem

At least 10 different naming conventions were used to create backup copies across the schema. This is why automated detection caught only 84 date-stamped tables through regex — a manual review pass was necessary to identify the full population of backup artifacts.

S.no	Style	Example
1	BKP prefix	BKP_MKTCASE1312025
2	_BAK suffix	MKTEMERGENCYREGIONDEF_BAK
3	_BK suffix	MKTNONSPINBILATERAL_BK
4	_BKUP suffix	MKTPLANDEADPNODEREPLACE_BKUP
5	_OLD suffix	MKTPLANCONSTRAINT_OLD
6	Date appended — MMDDYYYY	MKTCASE_05042020
7	Date appended — DDMMYYYY	BKP_MKTCASEBRANCH_25_FEB_2026
8	Date appended — MMDDYY	MKTAUTHORIZATION_1115
9	Date appended — YYYYMMDD	MKTASINTERFACEDF_20130115
10	Word date appended	BKP_MKTREASON_OCT232025

2.5 Empty vs Populated Tables

Table with data-(Num_rows>0) : 660

Empty tables- (Num_rows<0) : 424

S.no	Group	Total Tables	Empty
1	SNP — Snapshot	26	26
2	MIGRATION	4	4
3	AUDMKT — Audit	80	70
4	LPC — Core App	25	24
5	STG — Staging	52	50
6	MKT — Market	672	258
7	BKP — Backup	15	5

Finding 1 — SNP and MIGRATION groups are 100% empty. All 26 snapshot tables and all 4 migration tables have zero rows. These groups appear to be created that was set up but never populated, or was populated, used and then cleared. Either way they represent 30 tables with no current data value.

Finding 2 — 258 MKT tables are empty. Even within the core production group tables are empty. This means 258 of the 672 MKT tables exist in the schema as definitions — with indexes, constraints and backup overhead — but store no data.

Finding 3 — 70 of 80 AUDMKT audit tables are empty. The audit framework was built for 80 tables but only 10 are receiving audit records. The remaining 70 audit tables are defined but hold no audit trail.

2.6 Tables With No Date Columns — 116 Tables

During the data age analysis phase, every table was checked for the presence of CREATEDDATE or MODIFIEDDATE columns.

This check revealed that 116 tables have no date or timestamp columns at all.

Tables WITH date columns: 968

Tables WITHOUT any date columns: 116

Some tables have different names

CREATIONTIME, HISTCREATEDDATE, HISTMODIFIEDDATE, BATCHCREATEDDATE, BATCHMODIFIEDDATE, LASTUPDATE, CREATEDDATE, MODIFIEDDATE

Which groups have no date columns:

AQ\$ Oracle Internal, SNP Snapshot, MIGRATION, LPC internal, STG farm tables. Oracle system

For that we used the below for the age analysis.

DBA_TABLES.LAST_ANALYZED — when Oracle last gathered statistics

DBA_TAB_MODIFICATIONS.TIMESTAMP — when DML was last tracked.

Section 3: Storage Distribution

3.1 Storage by Group

s.NO	Group	Tables	Size
1	MKT — Market Tables	672	16,287.89
2	STG — Staging	52	59.93
3	OTHER — Unclassified	14	24.93
4	EMS System	6	21.41
5	Date-Stamped Backups	84	12.09
6	OLD Suffix Tables	6	11.27
7	BKP — Backup Tables	15	2.45
8	OPS — Operations	2	0.91
9	AUDMKT — Audit	80	0.86
10	TEMP Tables	10	0.75
11	LPC — Core App	25	0.43
12	GenByFuel	3	0.24
13	Equipment / Outage	3	0.19
14	MV Materialized Views	2	0.16
15	SPIN System	9	0.06
16	REGLOC Regulatory	2	0.02
17	MSCS System	5	0.01
18	All Remaining Groups	88	0.05
	TOTAL TABLE STORAGE	1084	16422.65

3.2 Top 10 vs Bottom 10 storage

Top-10 distribution according to the storage

s.no	table	segment type	size	rows
1	MKTPLANPNODE	TABLE PARTITION	5765.01	41 Billion
2	MKTPLANPNODECONSTRAINT_ARCH	TABLE PARTITION	2511.57	56.9 Billion
3	MKTPLANUNITSCHEDULE	TABLE	1308.2	1.5 Billion
4	MKTPLANPNODECONSTRAINT_D	TABLE PARTITION	1212.28	23.1 Billion
5	MKTPLANPNODE_D	TABLE PARTITION	1122.69	11.9 Billion
6	MKTPLANUNITASTYPE	TABLE PARTITION	929.92	6.9 Billion
7	MKTPLANPNODECONSTRAINT_ACTIVE	TABLE PARTITION	691.77	12.6 Billion
8	MKTPLANDEADPNODEREPLACE	TABLE	583.89	6.6 Billion
9	MKTPLANPNODE_P	TABLE PARTITION	374.01	4 Billion
10	MKTPLANPNODECONSTRAINT_P	TABLE PARTITION	284.44	5.5 Billion

Bottom-10 distribution according to the storage

S.no	Group	Total Tables	Empty
1	SNP — Snapshot	26	26
2	MIGRATION	4	4
3	AUDMKT — Audit	80	70
4	LPC — Core App	25	24
5	STG — Staging	52	50
6	MKT — Market	672	258
7	BKP — Backup	15	5

Section 4: Usage & Activity Metrics

Two views are used for the determination of table usage.

DBA_TAB_MODIFICATIONS - Tracks INSERT, UPDATE, DELETE counts and timestamp since last stats gathered

DBA_TABLES.LAST_ANALYZED - When Oracle last gathered statistics on the table

ACTIVEDML : detected within last 6 months

RECENTDML : detected 6–12 months ago

AGING : Last DML 1–5 years ago

STALE : Last activity 2016–5 years ago

PRE-2016 : Last activity before January 2016

4.1 Activity by group

Group	Active	Recent	Aging	Stale	Pre-2016	Empty	Total
MKT — Market	205	7	79	181	0	258	730
AUDMKT — Audit	7	0	4	0	0	70	81
STG — Staging	0	0	1	1	0	50	52
EMS System	1	0	1	5	0	0	7
LPC — Core App	2	0	0	6	0	24	32
BKP — Backup	2	2	6	0	0	5	15
SNP — Snapshot	0	0	0	0	0	26	26
ETM — Migration	0	1	0	11	0	6	18
MIGRATION	0	0	0	0	0	4	4
SPIN System	5	0	0	3	0	1	9
OPS — Operations	2	0	2	0	0	0	4
DATE-STAMPED	0	0	0	3	0	0	3
OTHER	11	1	0	7	2	24	45
MSCS System	2	0	0	1	0	1	4
AUD — Audit	1	0	0	0	0	1	2
All Remaining	1	0	0	2	0	0	3

4.2 Top and Bottom usage metrics

Top 10 — High Activity

These are the tables receiving the highest volume of data changes in the schema right now.

S.no	Table	Row Count	Inserts Tracked	Updated Tracked	Deleted Tracked	Last Activity
1	MKTPLANUNITASTYPE	6.9 Billion	3.9 Billion	400,199	3.1 Billion	22-Apr-26
2	MKTPLANUNITSCHEDULE	1.5 Billion	2.1 Billion	1.6 Billion	4.5 Million	22-Apr-26
3	MKTPLANPNODE	41 Billion	3.1 Billion	0	5.7 Million	22-Apr-26
4	MKTPLANPNODECONSTRAINT_D	23.1 Billion	232 Million	0	235,263	22-Apr-26
5	MKTPLANPNODECONSTRAINT_ACTIVE	12.6 Billion	239 Million	0	0	22-Apr-26
6	MKTPLANEADPNODEREPLACE	6.6 Billion	89 Million	0	43,013	22-Apr-26
7	MKTPNODEAGGRLMPHOURLY	2.4 Billion	131 Million	0	13.5 Million	13-Oct-25
8	MKTCOPRESOURCE	1.7 Billion	57 Million	0	34.7 Million	26-May-20
9	EMSZONALDISPATCHRATE	332 Million	30 Million	0	0	30-Jul-25
10	MKTPLANPAR	284 Million	20 Million	0	28,235	10-May-25

Bottom 10 — Lowest Activity (With Data, Not Empty)

These are the 10 tables that have rows but show the least write activity — either zero DML tracked or last activity the furthest in the past.

S.no	Table	Row Count	Last Activity	Inserts	Updated	Deleted	Inactive
1	AQS_ETMWINSRV_QT_S	1	22-Dec-14	2	0	0	11+ years
2	AQS_MARKET_QT_S	7	22-Dec-14	5	0	0	11+ years
3	AQS_SP_QT_S	1	22-Dec-14	1	0	0	11+ years
4	WSCSERVICE	1	19-Jul-12	0	0	0	13+ years
5	MKTPLANPNODECONSTRAINT_ARCH	56.9 Billion	1-Nov-12	0	0	0	13+ years
6	MKTCOPRESOURCE*	1.7 Billion	21-Oct-12	0	0	0	13+ years
7	MKTPLANZONEPNODELMP_OLD	90.6 Million	25-Oct-13	0	0	0	12+ years
8	MKTVALIDATIONMESSAGE_09302016	50.8 Million	30-Apr-13	0	0	0	13+ years
9	MKTPLANCONSTRAINT_OLD	22.5 Million	11-May-15	0	0	0	10+ years
10	MKTCOPCONSTRAINT	6 Million	19-Jun-14	0	0	0	11+ years

4.3 Tables Confirmed Dead — Entire Groups With Zero Activity

SNP-Snapshot : 26 tables - 0 rows - Last Activity is on 27 Jan 2017

Migration : 4 tables - 0 rows - Last Activity is on 27 Jan 2017

Both groups share the exact same last analyzed date — 27-JAN-2017.

4.4 Most critical examples — large active tables with stale statistics:

Table	Size GB	Last Analyzed	Years Stale
MKTPLANUNITSCHEDULE	1,308 GB	16-Feb-19	7 years
MKTPLANUNITASTYPE	929 GB	10-Jun-19	6 years
MKTCOPRESOURCE	140 GB	20-Apr-20	6 years
MKTPLANPNODE	5,765 GB	22-Feb-25	1 year

Findings:

A large number of tables across multiple groups share the same

SNP — All 26 tables

MIGRATION: All 4 tables

STG — Staging: 48 of 52 tables

LPC — Core App: 22 of 32 tables

ETM — Migration: 11 of 18 tables

MKT — Market (stale subset): ~150 tables

AQS — Oracle Internal: Multiple tables

Tables with last activity on 02-SEP-2021 (inactive since then):

MKTPLANNODE_P (374.01 GB)

MKTPLANNODECONSTRAINT_P (284.44 GB)

MKTPLANDISPATCH_P (0.42 GB)

MKTPLANPRD_P (1.33 GB)

Tables with later but still outdated activity (2022):

MKTPLANLRASTYPE_P → 04-OCT-2022

MKTPLANLRSERVICE_P → 23-SEP-2022

MKTPLANLOCALEPRODUCT_P → 22-SEP-2022

Tables still active (2025 activity):

MKTPLANLOCALE_P → 22-JUL-2025 (still active)

MKTPLANUNITASTYPE_P → 22-NOV-2025 (still active)

MKTPLANUNITSCHEDULE_P → 25-OCT-2025 (still active)

Section 5: Content & Metadata Analysis

5.1 Capturing comments- Initial analysis for the description of tables

Three Oracle system views were used to build the metadata picture:

DBA_TAB_COMMENTS : Free-text description of each table's purpose — added by developers at creation time

DBA_TAB_COLUMNS: Every column in every table — name, data type, position, nullable

DBA_COL_COMMENTS: Free-text description of individual columns — added by developers

Table with comments - 420

Table without comments - 664

5.2 Top 10 widest columns by column count

Table	Columns	Description
MKTCASE	589	Case management — largest column count in schema
MKTPLANUNITSCHEDULE	545	Unit schedule solution output
MKTPLANCONSTRAINT	479	Constraint solution output
MKTPLANCONSTRAINT_OLD	620	Old copy of constraint table — wider than current
MKTPLANLOCALE_OLD	461	Old copy of locale table
MKTPLANCONSTRAINT_D	373	Delta copy of constraint table
MKTPLANPNODE	226	PNode pricing solution
MKTPLANUNITIR	219	Unit IR solution
MKTPLANUNITSERVICE	203	Unit service solution
MKTPLANUNITASTYPE	201	Unit AS type solution

Finding:

The analysis shows that **MKTPLANCONSTRAINT_OLD** contains 620 columns, while the current **MKTPLANCONSTRAINT** table has only 479 columns, indicating that the old copy was taken when the schema was wider and before columns were later removed as part of schema. **MKTCASE** stands out with 589 columns, making it the widest table in the schema; such a high column count reflects continuous accumulation of requirements over time and makes the table harder to maintain, query, and index efficiently.

5.3 Table Definitions — Extracted from comments

1. MKTPLANPNODE

"This entity is used to hold the case solution PNODE price calculations."

Columns: 226

Size: 5,765 GB

Rows: 41 Billion

2. MKTCASE

"This entity is used to describe a Multi-Settlement Market study case. A case is used by a single user to study the day ahead market or supplementary commitment based on a set of configuration. The per case configuration attributes are stored in this entity while a finer grain per hour configuration is stored in MktCaseControl."

Columns: 589

Size: 0.89 GB

Rows: 3.3 Million

3. MKTPNODEPNODEHOURLY

"This table holds hourly factor data for pnode and its aggregate. This table is partitioned by day on MKTHOUR. Partitions are maintained by PJM."

Columns: 125

Size: 123 GB

Rows: 858 Million

4. EMSZONALDISPATCHRATE

"This entity contains the EMS zonal electronic rates used to dispatch units within the PJM control area. The information comes from Drate.txt file Start Rate Data — End Rate Data section."

Columns: 88

Size: 21.4 GB

Rows: 332 Million

5. MKTPLANPNODECONSTRAINT_ARCH

"This record contains the congestion component of each Pnode to binding constraint. This table is only written out by DSPD in LPC mode."

Section 6: Data Age & Archival Analysis

How much data exists in MOD_LPC that is older than January 2016 and has not been modified in the last 10 years?

Used CREATEDDATE and MODIFIEDDATE columns.

6.1 The following tables were identified as likely pre-2016 candidates

S.no	Table	Group	Size	Roes	Old record	Last modified	Years unmodified
1	MKTPLANPNODECONSTRAINT_ARCH	MKT-MKTPLAN	2511.57	56.9 Billion	30-Oct-12	1-Nov-12	13
2	MKTCOPRESOURCE	MKT-MKTCOP	140.11	1.7 Billion	13-Oct-12	21-Oct-12	13
3	MKTVALIDATIONMESSAGE_09302016	MKT-DATE-STAMPED	7.81	50.8 Million	6-Oct-12	30-Apr-13	13
4	MKTPLANZONEPNODELMP_OLD	MKT-MKTPLAN	4.88	90.6 Million	7-Aug-12	25-Oct-13	12
5	MKTPLANCONSTRAINT_OLD	MKT-MKTPLAN	4.86	22.5 Million	16-Sep-12	11-May-15	10
6	MKTCOPCONSTRAINT	MKT-MKTCOP	0.37	6 Million	13-Oct-12	19-Jun-14	11

Findings:

- Total GB: 2,669.60 GB
- Total Rows: ~58.8 Billion
- Number of tables: 6 tables
- Largest contributor: MKTPLANPNODECONSTRAINT_ARCH (2,511.57 GB — 94.1% of the total)
- Oldest data found: 07-AUG-2012
- Estimated annual storage cost: ~\$5,200 at \$2,000/TB

All 6 tables belong to the MKT group.

Executive Summary

Section 1 — Schema at a Glance

- Schema: MOD_LPC — Database: MOD_PROD
- Total size: 35,328 GB (34.5 TB) — 18% larger than prior 30TB estimate
- Total segments: 2,201 — includes everything (tables, indexes, partitions)
- Annual storage cost: ~\$70,600 at \$2,000/TB
- Total objects: ~20,925 across 17 types
- Largest object types: Index Partitions (7,909), Table Partitions (7,859), Indexes (1,699), Tables (1,084)
- Partitioned tables: 56 — generating 7,859 partition segments
- Non-partitioned tables: 1,028
- Largest partitioned table: MKTPLANPNODE — 2,453 partitions
- Backup takes 7 days because: 7,859 partitions + 1,699 indexes = 9,558 segments processed individually
- LOB columns: 22 across 9 tables — all consuming 0 GB

Section 2 — What Tables Exist

- Total tables: 1,084
- Tables with data: ~660
- Empty tables: ~424 (43%)
- Tables with no date columns: 116 — cannot be age-assessed
- Tables with no documentation: 664 (61%)
- Classification approach: prefix analysis + suffix analysis + manual review
- 34 functional groups identified — MKT (672) is the largest
- 10 different backup naming conventions — BKP_ prefix, _BAK, _BK, _BKUP, _OLD, and 5 date formats
- SNP (26 tables) and MIGRATION (4 tables) — 100% empty, confirmed dead groups
- 258 MKT tables are empty — exist as definitions only, no data
- 70 of 80 AUDMKT audit tables are empty — audit framework built but not activated

Section 3 — Where the Storage Is

- MKT group: 16,287 GB — 46% of entire schema
- Total table storage: 16,422 GB
- Remaining ~18,900 GB is index segments
- 96% of table storage sits in just 56 partitioned tables
- Top 10 tables = 41.8% of entire schema
- MKTPLANPNODE family (7 tables) = 11,961 GB = 33.9% of schema
- Single largest table: MKTPLANPNODE — 5,765 GB, 41 billion rows
- Second largest: MKTPLANPNODECONSTRAINT_ARCH — 2,511 GB, 56.9 billion rows
- Bottom 10 smallest populated tables — all 0.0001 GB — singleton reference tables (1–3 rows each)
- Size range across schema: 57 million to 1 ratio — largest to smallest populated table

Section 4 — Usage of Tables

How usage was measured:

- DBA_TAB_MODIFICATIONS — tracks INSERT, UPDATE, DELETE and timestamp
- DBA_TABLES.LAST_ANALYZED — fallback when DML data is missing
- Combined using NVL logic — DML timestamp first, last analyzed as fallback

%ACTIVE — within 6 months 241 22%

RECENT — 6 to 12 months 11 1%

AGING — 1 to 5 years 91 8%

STALE — 2016 to 5 years 219 20%

EMPTY — zero rows 464 43%

Section 6 — Pre-2016 Data Age Analysis

Methodology — Three steps:

1. MIN(CREATEDDATE) < 01-JAN-2016 → data predates 2016
 2. MAX(MODIFIEDDATE) < 01-JAN-2016 → not modified in 10 years
 3. Collect size — both steps must pass
- All _DEF tables excluded per manager's instruction
 - Direct queries run on 5–10 million row samples using ROWNUM
 - Metadata alone was not sufficient — Oracle re-analyzes stats even on frozen tables

MKTPLANPNODECONSTRAINT_ARCH = 94.1% of the entire pre-2016 total

- All 6 belong to MKT group
- Oldest data: 07-AUG-2012

S.no	Table	Group	Size	Roes	Old record	Last modified	Years unmodified
1	MKTPLANPNODECONSTRAINT_ARCH	MKT-MKTPLAN	2511.57	56.9 Billion	30-Oct-12	1-Nov-12	13
2	MKTCOPRESOURCE	MKT-MKTCOP	140.11	1.7 Billion	13-Oct-12	21-Oct-12	13
3	MKTVALIDATIONMESSAGE_09302016	MKT-DATE-STAMPED	7.81	50.8 Million	6-Oct-12	30-Apr-13	13
4	MKTPLANZONEPNODELMP_OLD	MKT-MKTPLAN	4.88	90.6 Million	7-Aug-12	25-Oct-13	12
5	MKTPLANCONSTRAINT_OLD	MKT-MKTPLAN	4.86	22.5 Million	16-Sep-12	11-May-15	10
6	MKTCOPCONSTRAINT	MKT-MKTCOP	0.37	6 Million	13-Oct-12	19-Jun-14	11